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ITA-0329:Mobile Computing

Future Trends in Mobile Computing and Intelligent Cellular Networks

CO1-AT3-Individual Talk

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INTRODUCTION
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INTRODUCTION

- Future Trends: The shift toward technologies deploying between now and 2030.
- Mobile Computing: Powering data processing on the move for smartphones, autonomous vehicles, and robotics.
- Intelligent Cellular Networks: Wireless infrastructure (5G-Advanced/6G) using built-in AI to self-optimize and route data automatically.
- Future trends in mobile computing and intelligent cellular networks are shifting from traditional voice- and data-centric setups toward autonomous, AI-driven, and high-frequency architectures.

PROBLEM STATEMENT

- Network Overload: Legacy networks cannot handle the massive data explosion and heavy AI uploading from billions of connected smart devices.
- Wasted Efficiency: Older "dumb pipe" infrastructure relies on slow, manual human fixes during traffic spikes or unexpected outages.
- Blind Infrastructure: Current networks have no physical awareness of their surroundings, leading to dropped signals and poor spatial tracking.
- Massive Energy Drain: Blasting high-power wireless signals uniformly across entire cities is highly inefficient.

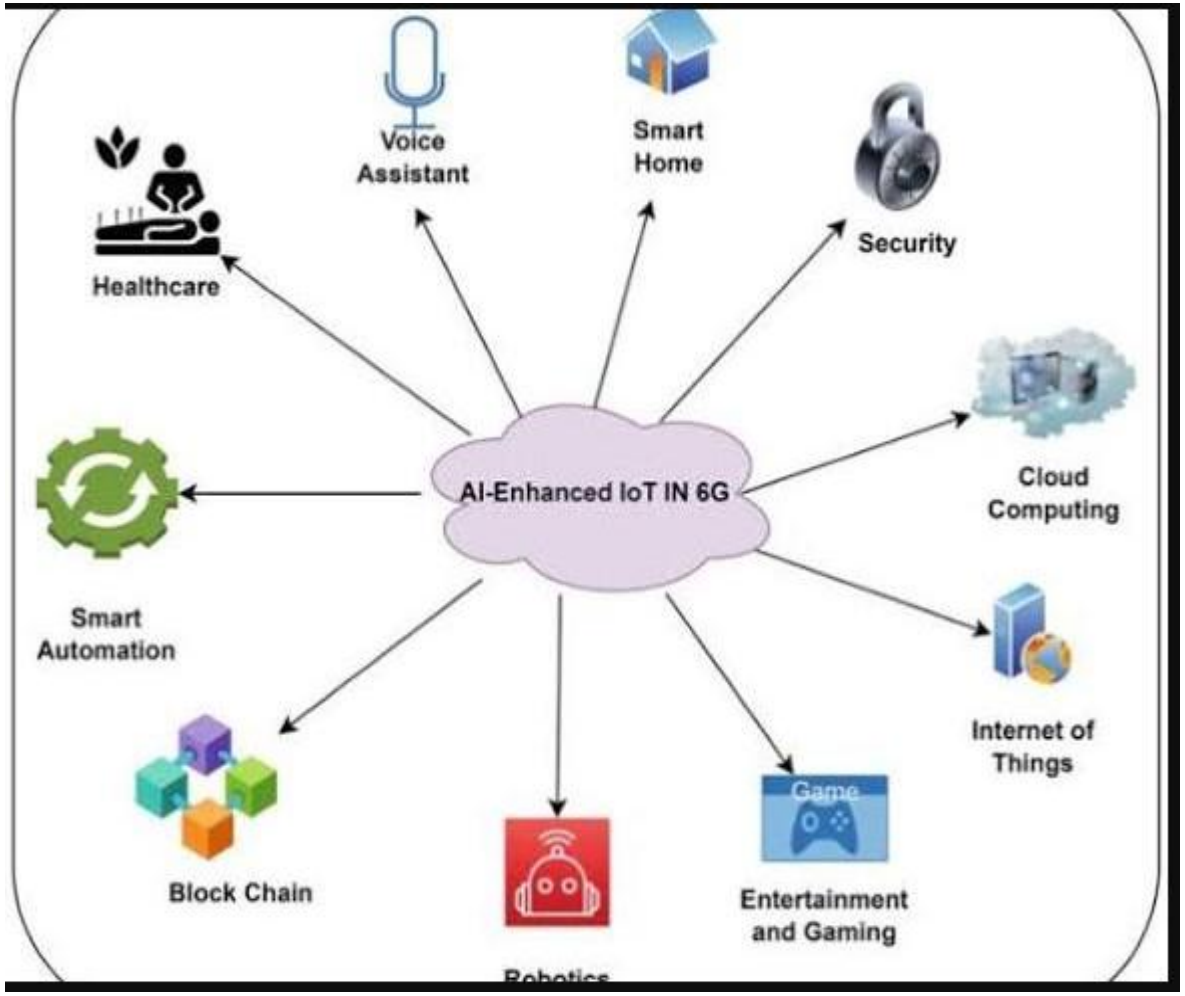
OBJECTIVES

- Maximize Bandwidth: Open up new high-frequency spectrums (like 6G terahertz) to cleanly handle massive data and heavy AI upload traffic.
- Automate Network Control: Build AI directly into the towers so the network can instantly self-optimize and fix outages without human help.
- Enable Environmental Awareness: Use the network signals like radar (ISAC) so infrastructure can track physical objects and drones in real time.
- Drastically Cut Energy Waste: Implement smart power-saving architectures that only direct high-power signals exactly where active devices are located.

KEY TECHNOLOGIES

- 6G Terahertz Spectrum: Ultra-high frequency bands that deliver extreme data rates and very low latency.
- AI-RAN: Machine learning built into radio towers to self-optimize, predict faults, and heal outages automatically.
- ISAC (Sensing + Communication): The same radio signals also work like radar to detect objects and drones.
- Smart Beamforming: Focuses power only toward active devices, which cuts energy waste across the network.

AI-Enhanced IoT in 6G



Project Overview

- The Goal: Build an "AI-driven digital fabric" that turns basic wireless towers into smart, self-thinking computing platforms.
- The Tech: Implements AI directly into radio towers (AI-RAN) so the network can self-heal, fix outages, and route data instantly.
- The Evolution: Upgrades legacy systems into software-driven 6G networks that effortlessly handle high-volume AI uploads from cars, robots, and drones.
- The Success Metric: Evaluated by achieving zero lag, maximizing ultra-high frequency speeds, tracking objects via radar-like sensing, and cutting power waste.

Challenges and Future Scope

- Signal Range: Terahertz waves are easily blocked by walls and weather, so dense small-cell coverage is needed.
- High Cost: Deploying AI-enabled towers and new spectrum hardware needs heavy investment.
- Security and Privacy: Radar-like sensing and AI control raise new data protection and cyber-attack risks.
- Future Scope: Self-driving vehicles, smart cities, and robotics will rely on 6G by 2030 and beyond.

Conclusion

- This project shifts cellular networks from simple pipelines into intelligent, self-operating compute platforms.
- Built-in AI-RAN and 6G eliminate data lag and allow the network to self-heal instantly.
- Radar-like environmental sensing bridges the gap between digital networks and physical AI like autonomous vehicles.
- Despite near-term signal and cost hurdles, this architecture is vital for a smart global ecosystem.

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